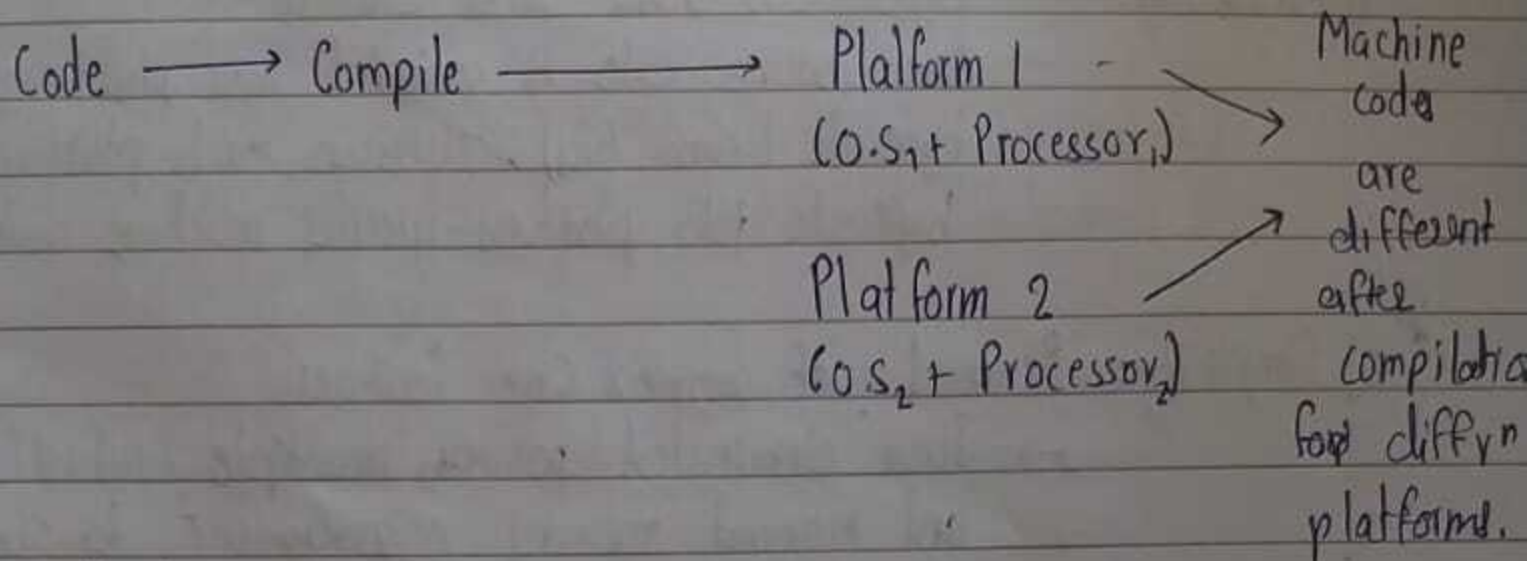


Introduction to Java

* Why was Java Created?

→ before java, C & C++ dominated, but it has Major problems

1] Portability issue



2] Complexity

→ features like pointers, manual memory allocation & multiple inheritance made C++ complex.

3] Security

→ C++ programs allowed unsafe operations, making systems vulnerable

* Platform

→ Platform is a combination of :
• Processor
• Operating system

- Each platform has a unique Instruction Set Architecture (ISA) which defines how software commands are executed at the processor level. This difference in ISA causes compiled binaries to be platform specific

* Java's Solution to these problems

- Java was designed to address the three main issues

1] Portability: Bytecode + JVM was solution

→ Java source code is compiled into platform-independent bytecode (.class files). JVMs on each platform interpret this bytecode into platform-specific machine code at runtime.

2] Simplicity: Removal of complex C++ features

→ Java eliminated pointers, multiple inheritance (via classes) and manual memory management to simplify programming.

3] Security: JVM's sandbox environment

→ JVM runs bytecode in restricted environment preventing unsafe operations, protecting the system from malicious code.

* Points to be Noted

→ While bytecode is independent of platform, JVM is platform dependent meaning a JVM must be implemented for each platform.

→ Java was used extensively both on the backend (via servlets) and the frontend (via applets, though applets are now deprecated).

→ Java follows WORA (Write once, ~~run~~ Run Anywhere)